CLOUD SECURITY SPECIALIST

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LAB:SERVICE ENDPOINTS AND SECURING STORAGE

1.1 INTRODUCTION

I started the lab with a required lab set up.I clicked on ctrl alt and delete to bring up the login page,signed in as admin and then downloaded the lab files.

PREPARE CLOUD SHELL FOR LATER USE

Step 1

I logged in to the Azure portal using the credentials provided by the lab.

Step 2

In the toolbar at the top of the Azure portal, I selected the Cloud Shell icon.

Step 3

In the Welcome to Azure Cloud Shell dialog, I selected PowerShell.

Step 4

On the Getting Started screen, I selected Mount Storage Account.

Step 5

In the Storage account subscription drop down, I selected my subscription then clicked on the apply button.

Step 6

In the Mount Storage Account screen,I selected I want to create a storage account.

Step 7

On the Create Storage Account screen, I filled the fields for Resource group,Storage account, File share and Region, then clicked on the Create button

Step 8

After the cloud shell initialized and put me at a text prompt, I exited the shell.

LAB 06:SERVICE ENDPOINTS AND SECURING STORAGE

LAB SCENARIO:

I was asked to create a proof of concept to demonstrate securing Azure file shares.

EXERCISE 1:SERVICE ENDPOINTS AND SECURITY STORAGE

In this exercise, I completed the following tasks:Create a virtual network,Add a subnet to the virtual network and configure a storage endpoint,Configure a network security group to restrict access to the subnet,Configure a network security group to allow rdp on the public subnet,Create a storage account with a file share,Deploy virtual machines into the designated subnets,Test the storage connection from the private subnet to confirm that access is allowed and Test the storage connection from the public subnet to confirm that access is denied.

TASK 1:CREATE A VIRTUAL NETWORK

In this task,I created a virtual network.

Step 1

I signed in to the Azure portal

Step 2

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Virtual networks and pressed the Enter key.

Step 3

On the Virtual Networks blade, I clicked on + Create.

Step 4

On the Basics tab of the Create virtual network blade, I specified the settings provided for;Subscription,Resource group,name and Region, left the others with their default values and clicked Next: IP

Step 5

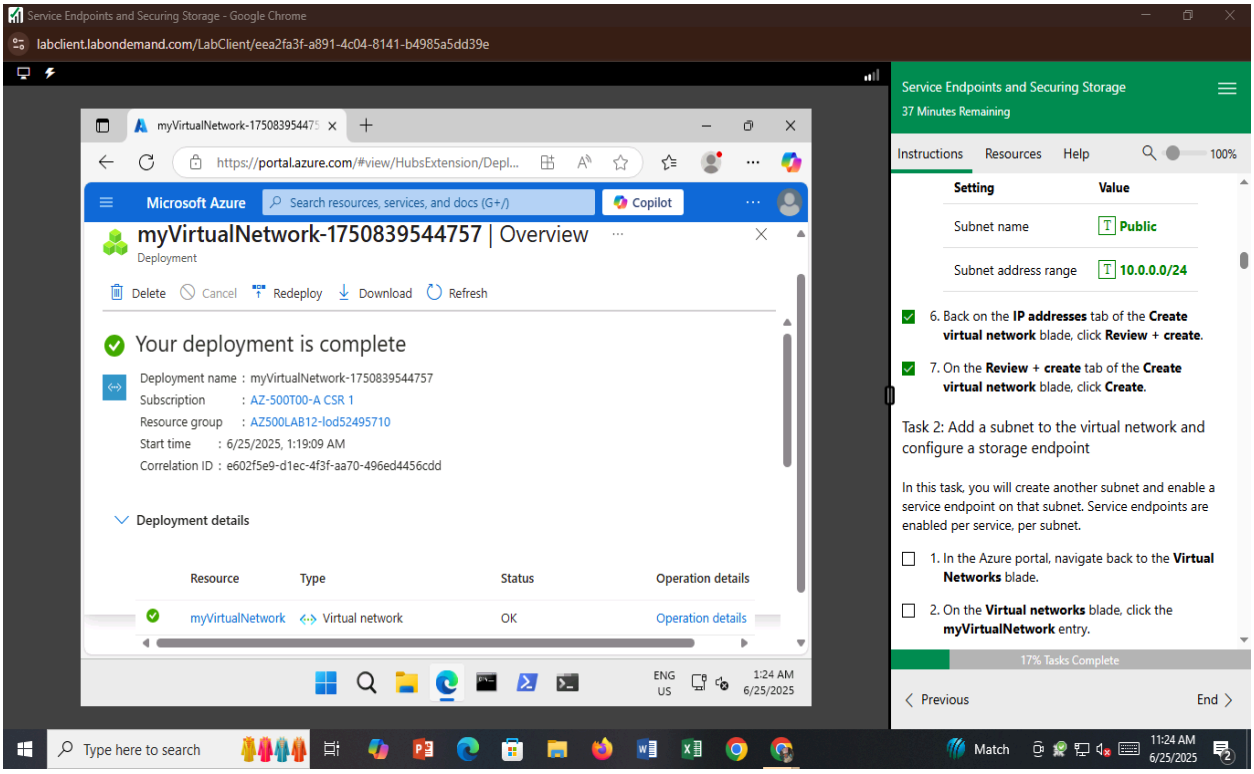
On the IP addresses tab of the Create virtual network blade, I set the IPv4 address space to 10.0.0.0/16, in the Subnet name column, clicked default and, on the Edit subnet blade, specified the settings provided for Subnet name and Subnet address range and clicked on Save.

Step 6

Back on the IP addresses tab of the Create virtual network blade, I clicked on the Review + create button.

Step 7

On the Review + create tab of the Create virtual network blade,I clicked the Create button to create the virtual network.Here is my screenshot



TASK 2:ADD A SUBNET TO THE VIRTUAL NETWORK AND CONFIGURE A STORAGE ENDPOINT

In this task, I created another subnet and enabled a service endpoint on that subnet. Service endpoints are enabled per service, per subnet.

Step 1

In the Azure portal, I navigated back to the Virtual Networks blade.

Step 2

On the Virtual networks blade, I clicked the myVirtualNetwork entry.

Step 3

On the myVirtualNetwork blade, in the Settings section, I clicked on Subnets.

Step 4

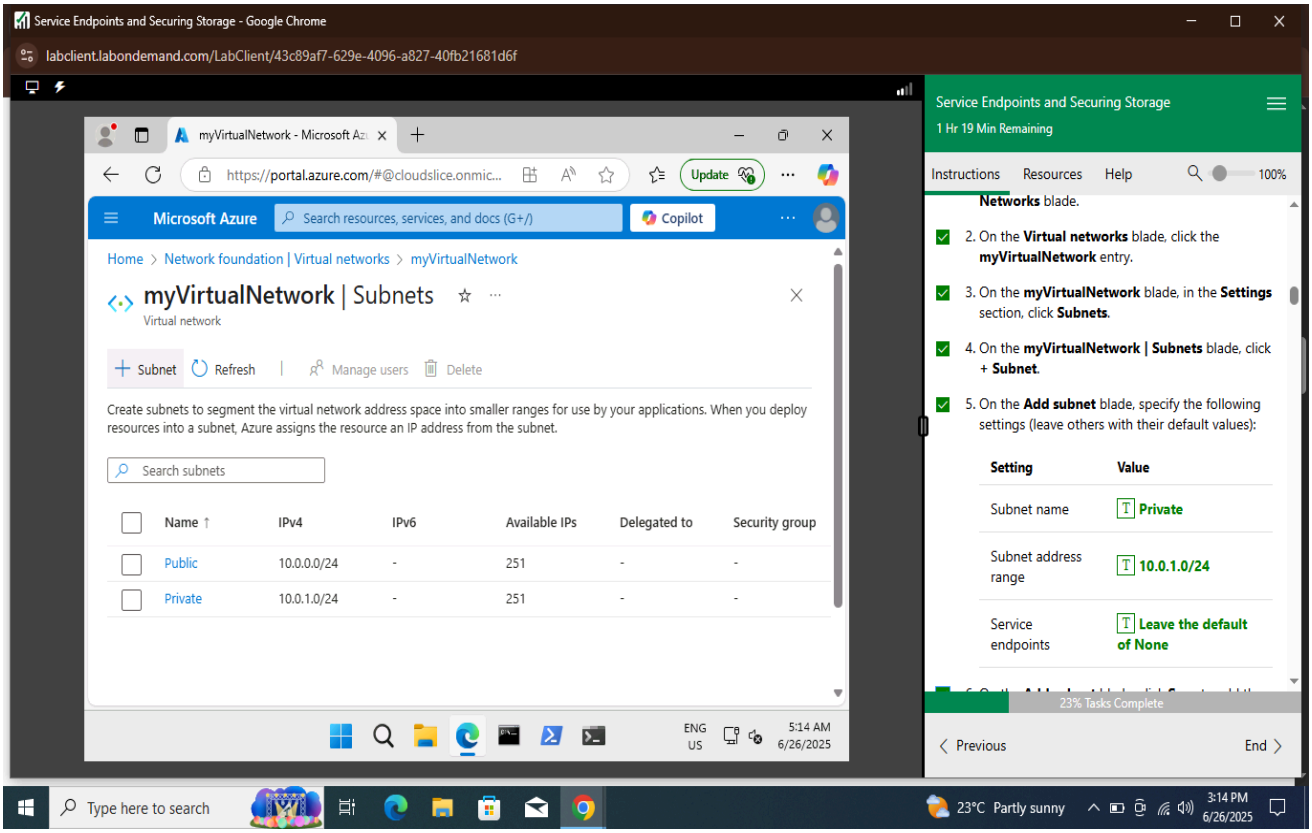
On the myVirtualNetwork | Subnet's blade, I clicked + Subnet.

Step 5

On the Add subnet blade, I specified the settings provided for subnet name,subnet address range and service endpoints and left others with their default values.

Step 6

On the Add subnet blade, click add to add the new subnet.The virtual network now has two subnets: Public and Private.Here is my screenshot



TASK 3:CONFIGURE A NETWORK SECURITY GROUP TO RESTRICT ACCESS TO THE SUBNET

In this task, I created a network security group with two outbound security rules (Storage and internet) and one inbound security rule (RDP). I also associated the network security group with the Private subnet. This will restrict outbound traffic from Azure VMs connected to that subnet.I skipped steps 1 to 8 because a workaround that automatically creates the myNsgPrivate Network Security Group and its first Outbound Security Rule was created.

Step 9

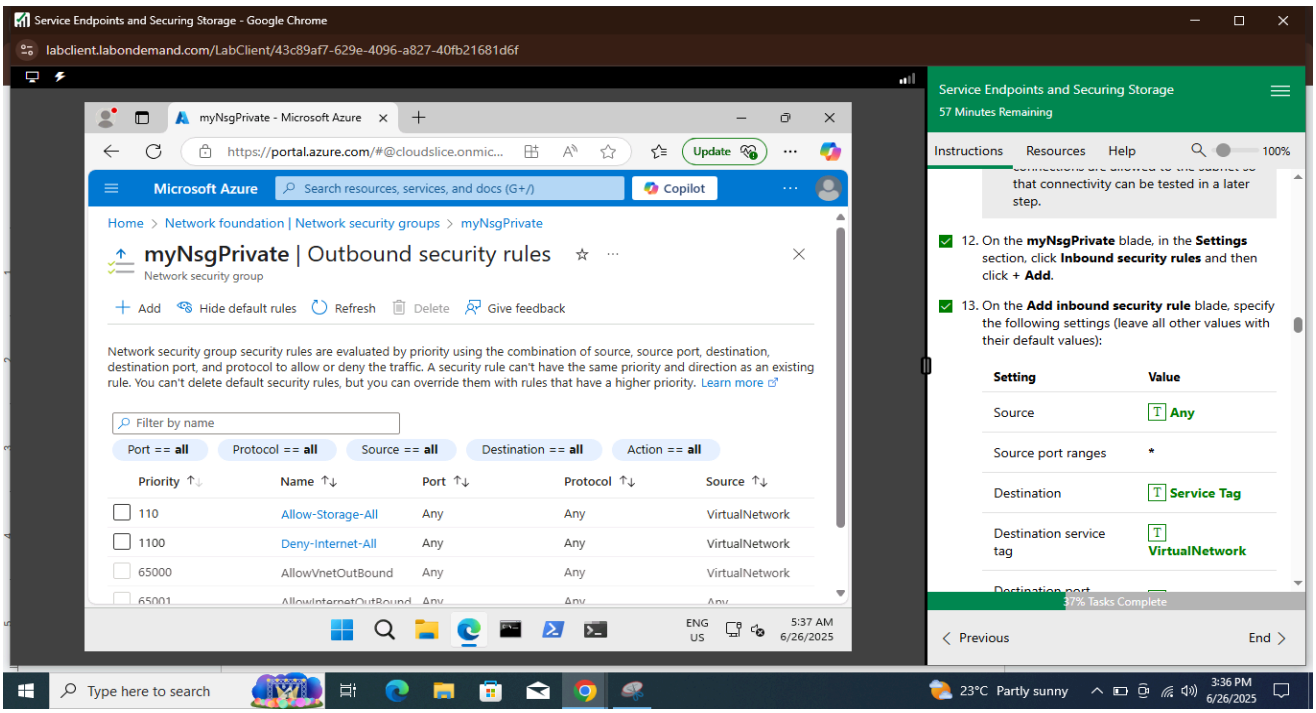
On the Add outbound security rule blade, I clicked Add to create the new outbound rule.

Step 10

On the myNsgPrivate blade, in the Settings section,I clicked Outbound security rules, and then + Add

Step 11

On the Add outbound security rule blade, I specified the settings provided to explicitly deny outbound traffic to the Internet and left all other values with their default settings.Here is my screenshot



Step 12

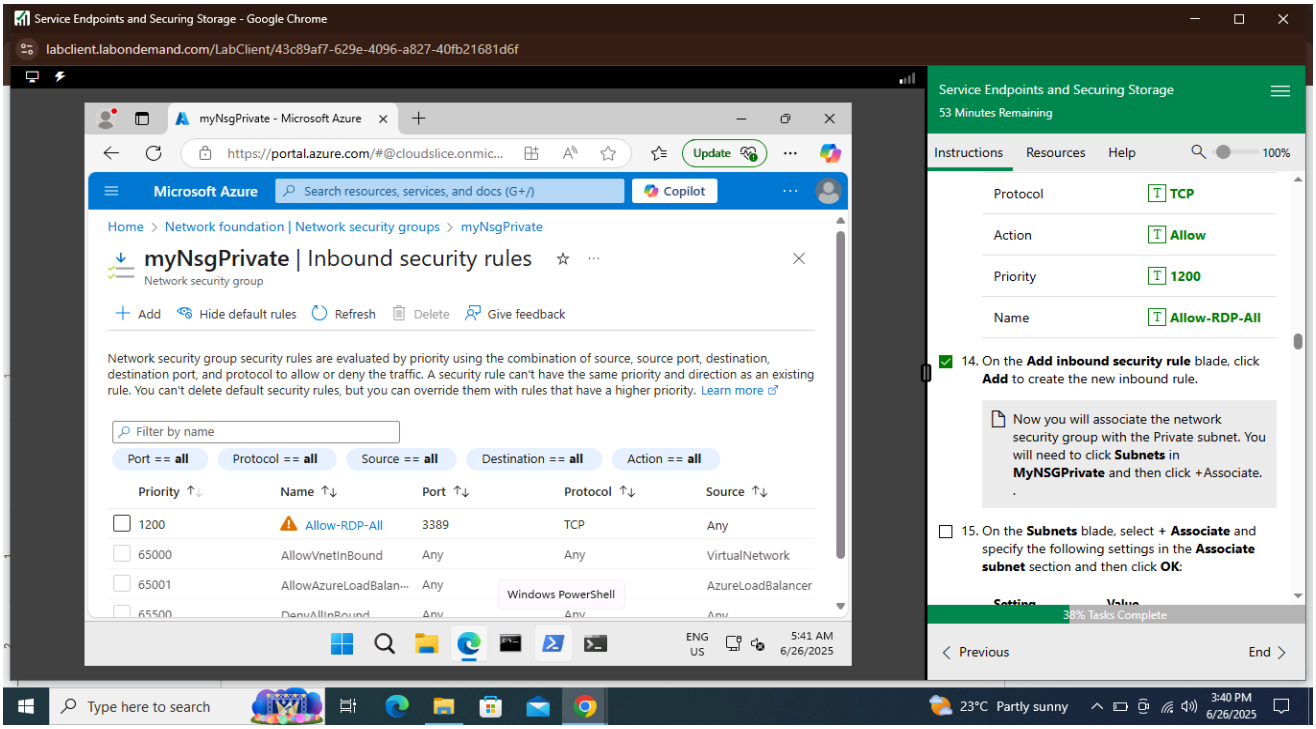
On the myNsgPrivate blade, in the Settings section, I clicked Inbound security rules and then + Add.

Step 13

On the Add inbound security rule blade, I specified the settings provided and left all other values with their default values.

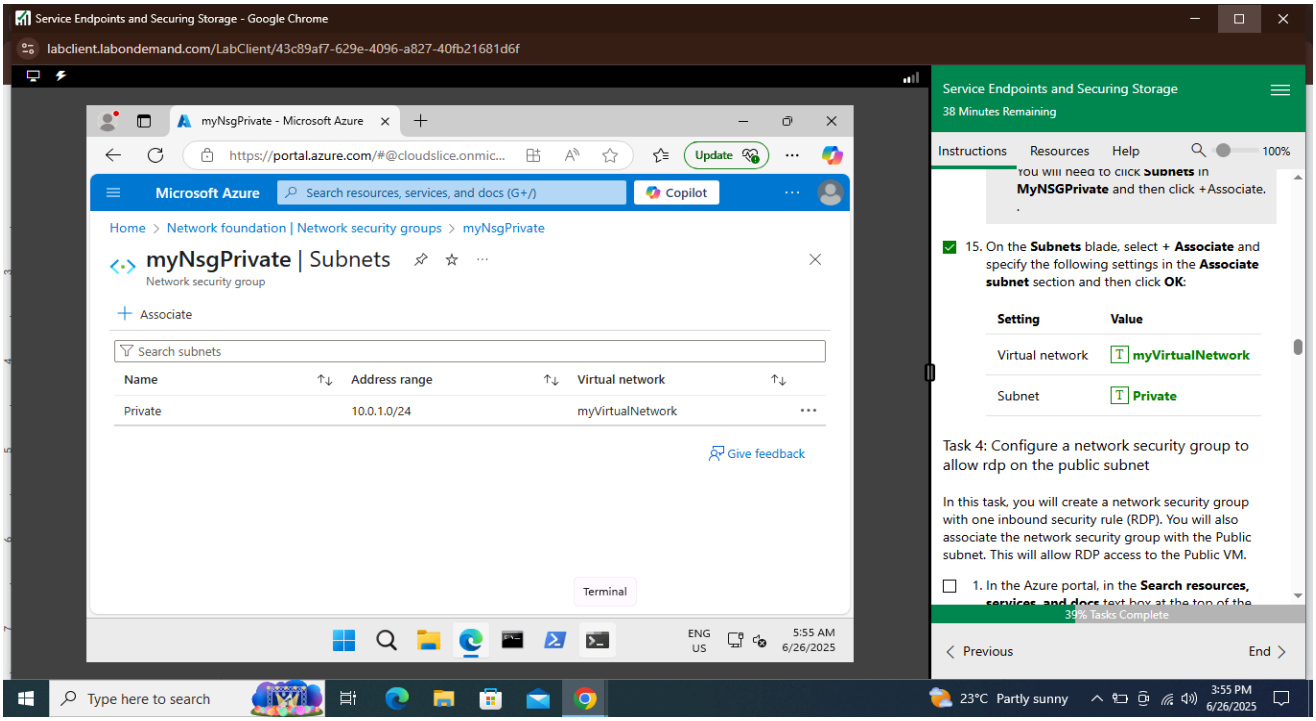
Step 14

On the Add inbound security rule blade, I clicked Add to create the new inbound rule.Here is my screenshot



Step 15

On the Subnets blade, I selected + Associate and specified the settings provided for virtual networks and subnet in the Associate subnet section and then clicked OK.Here is my screenshot



TASK 4:CONFIGURE A NETWORK SECURITY GROUP TO ALLOW RDP ON THE PUBLIC SUBNET

In this task, I created a network security group with one inbound security rule (RDP). Also associate the network security group with the Public subnet. This will allow RDP access to the Public VM.

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Network security groups and pressed the Enter key.

Step 2

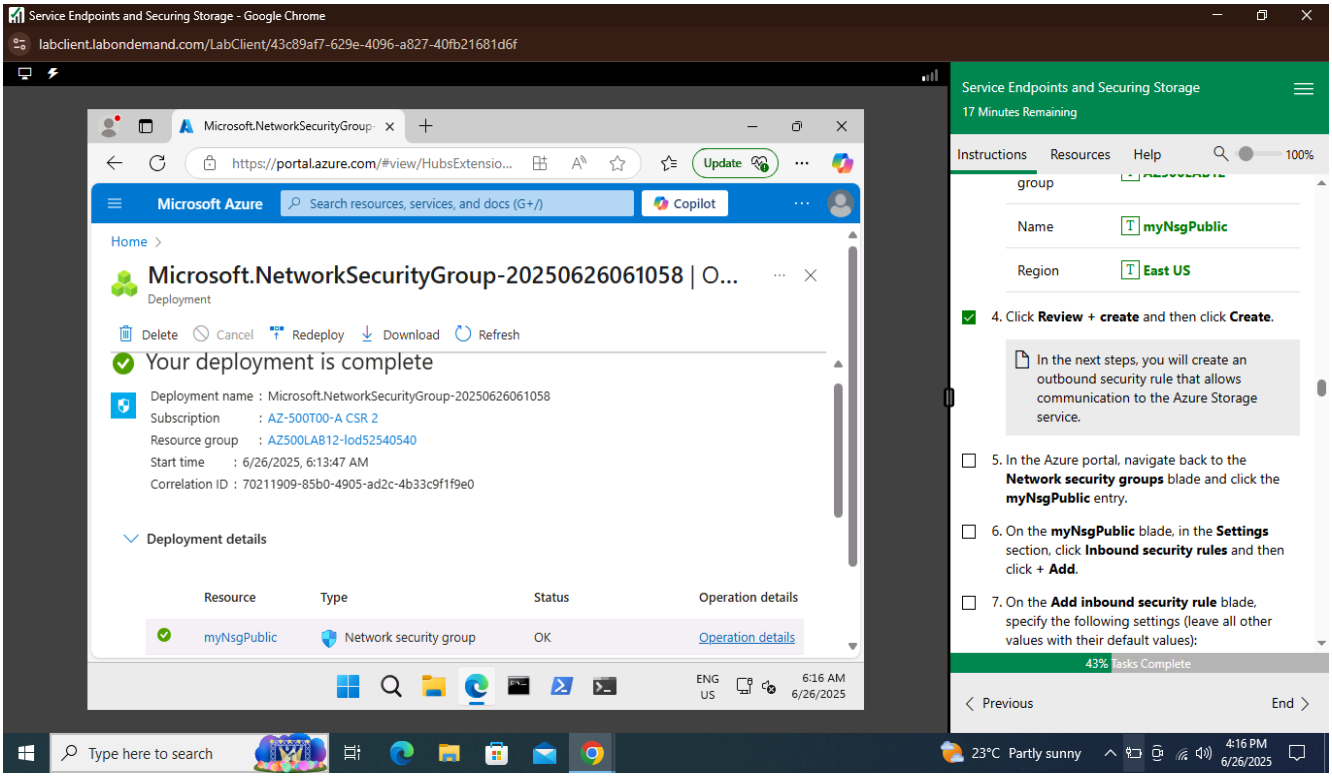
On the Network security groups blade, I clicked + Create.

Step 3

On the Basics tab of the Create network security group blade, I specified the settings provided.

Step 4

I Clicked on Review + create and then Create.Here is my screenshot



Step 5

In the Azure portal,I navigated back to the Network security groups blade and clicked the myNsgPublic entry.

Step 6

On the myNsgPublic blade, in the Settings section, I clicked on Inbound security rules and then clicked + Add.

Step 7

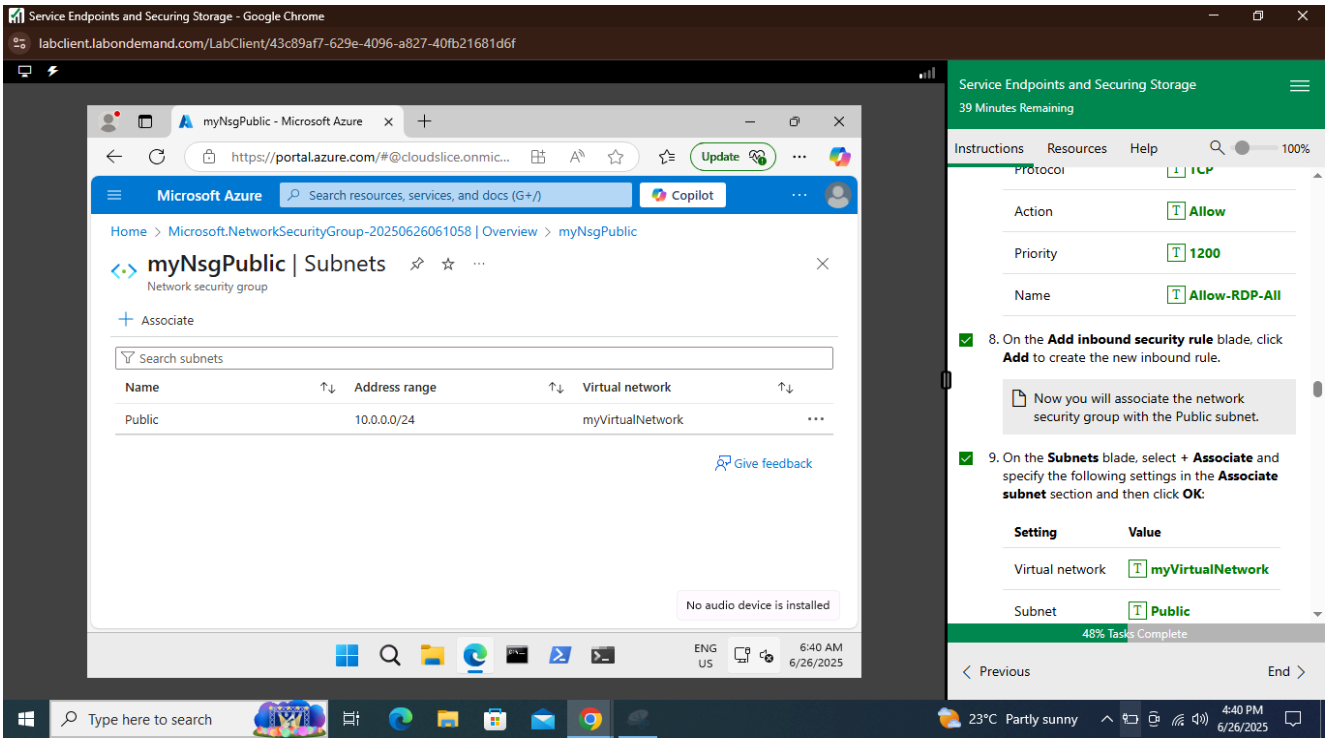
On the Add inbound security rule blade, I specified the settings provided and left all other values with their default values.

Step 8

On the Add inbound security rule blade, I clicked Add to create the new inbound rule

Step 9

On the Subnets blade, I selected + Associate and specified the settings provided for virtual network and subnet in the Associate subnet section and then clicked OK.Here is my screenshot



TASK 5:CREATE A STORAGE ACCOUNT WITH A FILE SHARE

In this task,I created a storage account with a file share and obtained the storage account key.

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Storage accounts and pressed the Enter key.

Step 2

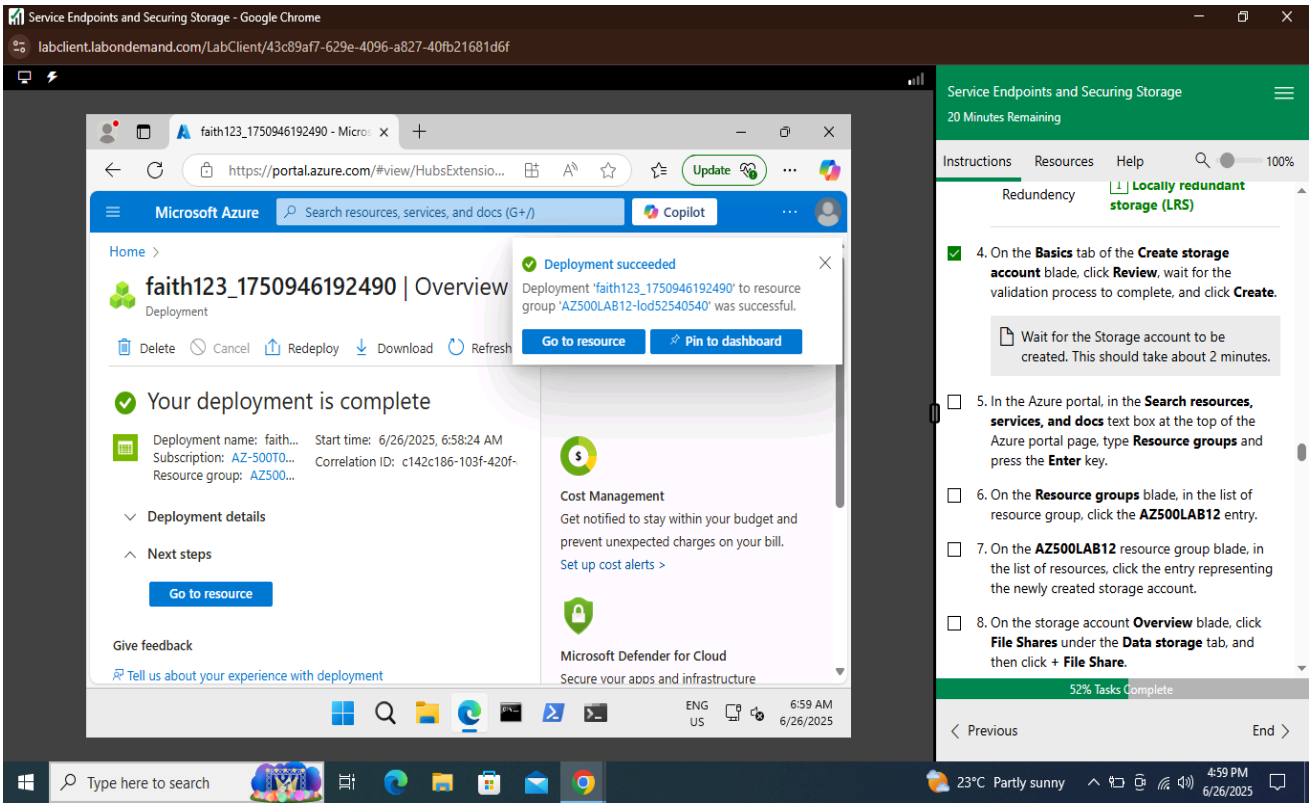
On the Storage accounts blade, I clicked + Create.

Step 3

On the Basics tab of the Create storage account blade, I specified the settings provided and left others with their default values.

Step 4

On the Basics tab of the Create storage account blade, I clicked on Review, waited for the validation process to complete, and clicked Create.Below is my screenshot



Step 5

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, I typed Resource groups and pressed the Enter key.

Step 6

On the Resource groups blade, in the list of resource group, I clicked the AZ500LAB12 entry.

Step 7

On the AZ500LAB12 resource group blade, in the list of resources, I clicked on the entry representing the newly created storage account.

Step 8

On the storage account Overview blade, I clicked File Shares under the Data storage tab, and then clicked + File Share.

Step 9

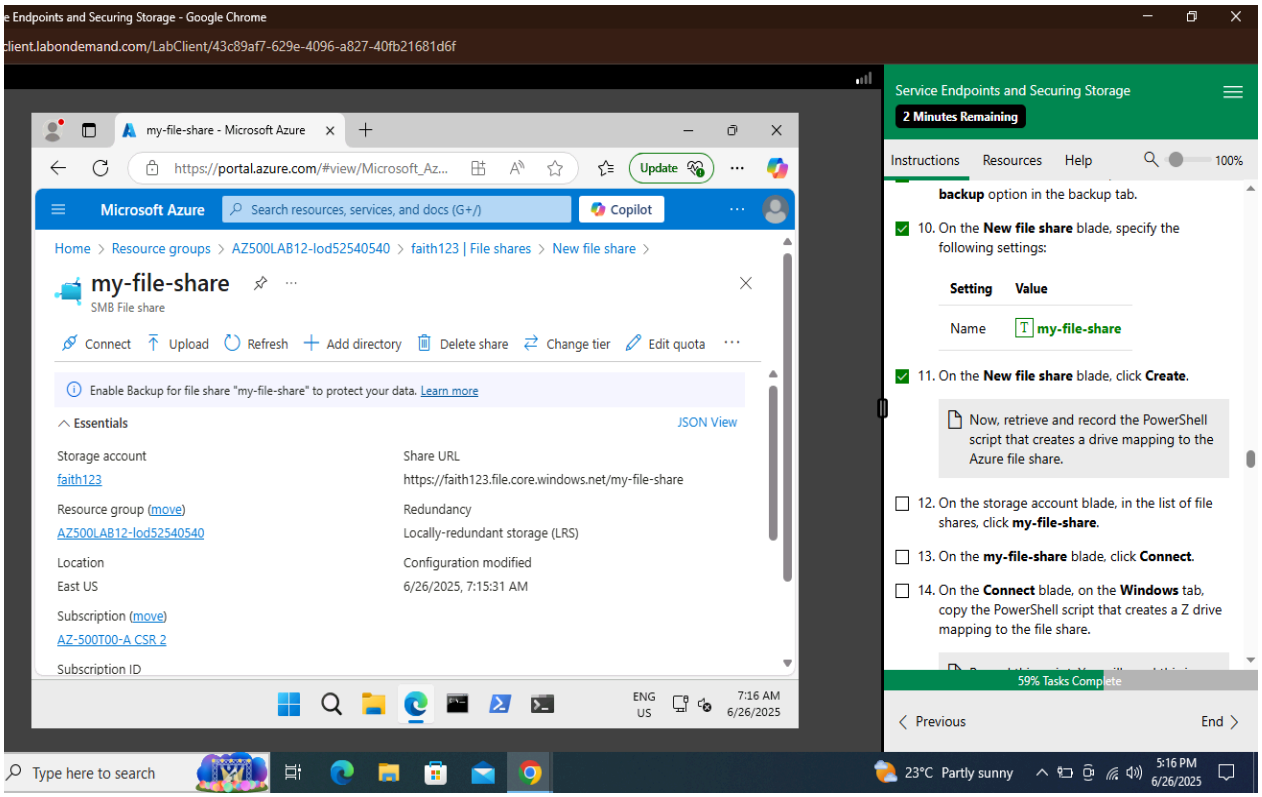
On the New file share blade, I unticked the Enable backup option in the backup tab.

Step 10

On the New file share blade,I entered the name proposed for file share name.

Step 11

On the New file share blade, click Create.Below is my screenshot



Step 12

On the storage account blade, in the list of file shares, I clicked my-file-share.

Step 13

On the my-file-share blade, I clicked Connect.

Step 14

On the Connect blade, on the Windows tab, I copied the PowerShell script that creates a Z drive mapping to the file share.

Step 15

I navigated back to the storage account blade, then in the Security + networking section, clicked on Networking.

Step 16

Under Public network access, I selected Manage and as Default action selected Enable from selected networks.

Step 17

On the Add networks blade,I specified the settings provided for;Subscription,virtual networks and subnets.Here is my screenshot

Step 18

On the Add networks blade, I specified the settings for:Subscription,virtual networks and subnets

Step 19

On the Add networks blade,I clicked Add

Step 20

Back on the storage account blade,I clicked on Save.At this point in the lab I had configured a virtual network, a network security group, and a storage account with a file share.

TASK 6:DEPLOY VIRTUAL MACHINES INTO A DESIGNATED SUBNETS

In this task, I created two virtual machines, one in the Private subnet and one in the Public subnet.

Step 1

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page,I typed Virtual machines and pressed the Enter key.

Step 2

On the Virtual machines blade, I clicked + Create and, in the dropdown list, clicked + Azure Virtual machine

Step 3

On the Basics tab of the Create a virtual machine blade, I specified the settings provided and left others with their default values

Step 4

I Clicked Next: Disks > and, on the Disks tab of the Create a virtual machine blade, set the OS disk type to Standard HDD and clicked Next: Networking >.

Step 5

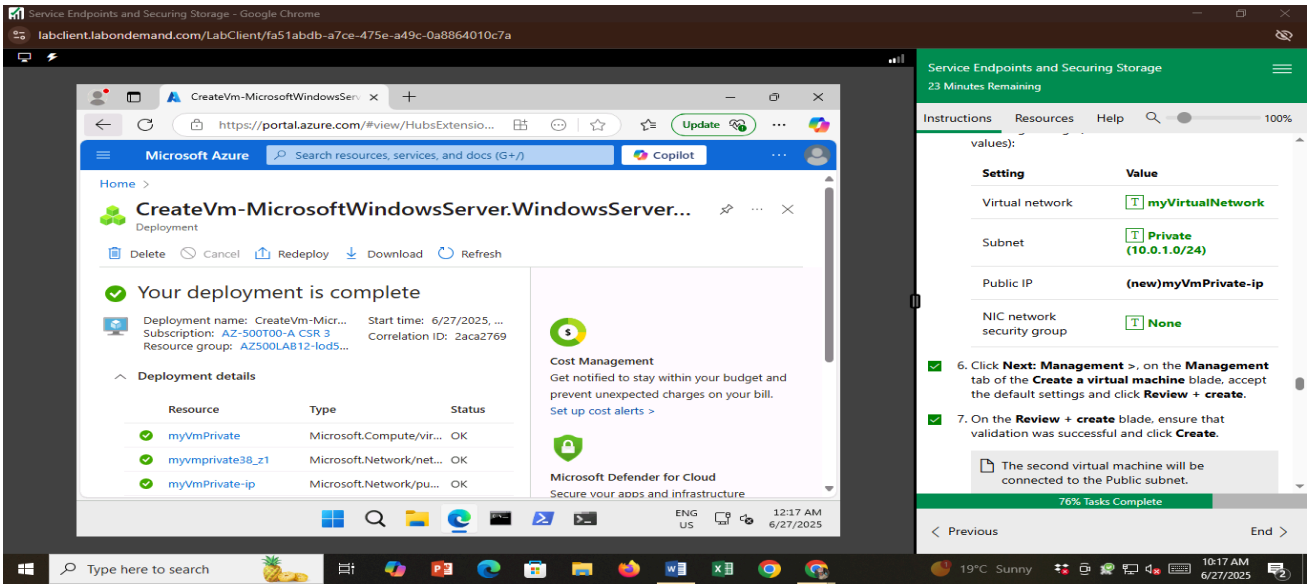
I Clicked Next: Networking >, on the Networking tab of the Create a virtual machine blade, specified the settings provided and left others with their default values.

Step 6

I Clicked Next: Management >, on the Management tab of the Create a virtual machine blade, accepted the default settings and clicked Review + create.

Step 7

On the Review + create blade, I ensured that validation was successful and clicked on the Create button.Here is my screenshot



Step 8

On the Virtual machines blade, I clicked +Create and, in the dropdown list, clicked + Azure Virtual machine.

Step 9

On the Basics tab of the Create a virtual machine blade, I specified the settings provided and left others with their default values.

Step 10

I Clicked Next: Disks > and, on the Disks tab of the Create a virtual machine blade, set the OS disk type to Standard HDD and clicked Next: Networking >.

Step 11

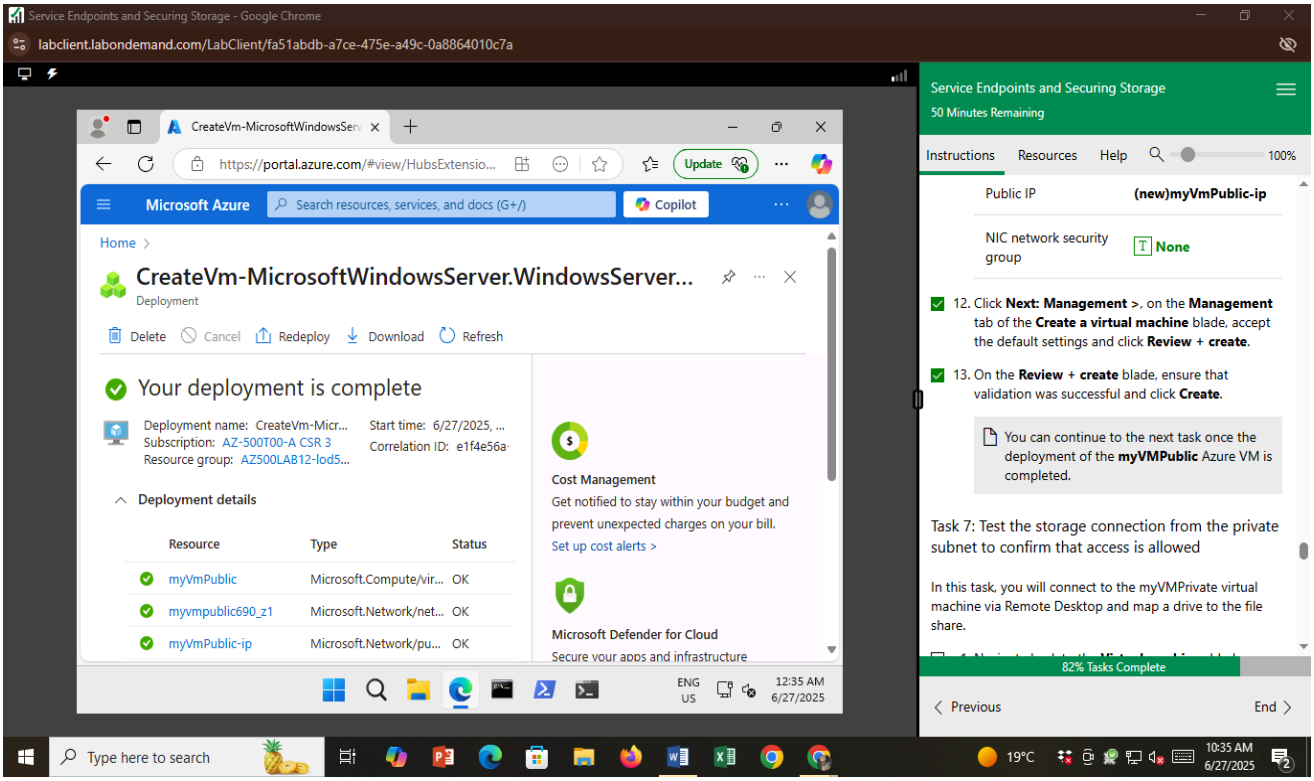
I Clicked Next: Networking >, on the Networking tab of the Create a virtual machine blade, specified the settings provided and left others with their default values.

Step 12

I clicked Next: Management >, on the Management tab of the Create a virtual machine blade, accepted the default settings and clicked Review + create.

Step 13

On the Review + create blade, I ensured that validation was successful and clicked Create.Here is my screenshot



TASK 7:TEST THE STORAGE CONNECTION FROM THE PRIVATE SUBNET TO CONFIRM THAT ACCESS IS ALLOWED

In this task,I connected to the myVMPrivate virtual machine via Remote Desktop and mapped a drive to the file share.

Step 1

I Navigated back to the Virtual machines blade.

Step 2

On the Virtual machines blade, I clicked the myVMPrivate entry.

Step 3

On the myVMPrivate blade, I clicked Connect and, in the drop down menu, clicked Connect.

Step 4

I downloaded the RDP file and used it to connect to the myVMPrivate Azure VM via Remote Desktop.

Step 5

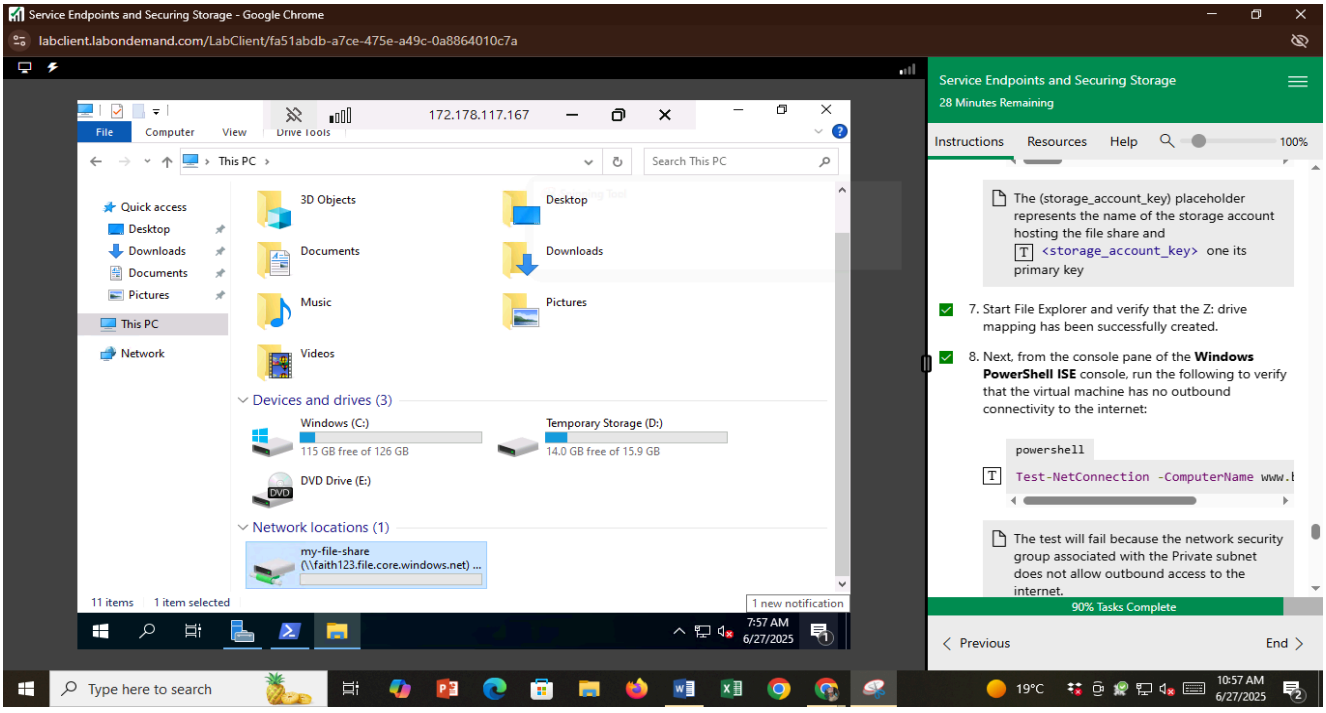
Within the Remote Desktop session to myVMPrivate, I clicked Start and then clicked Windows PowerShell ISE.

Step 6

Within the Windows PowerShell ISE window, I opened the Script pane, then pasted and ran the PowerShell script that I recorded earlier in the lab.

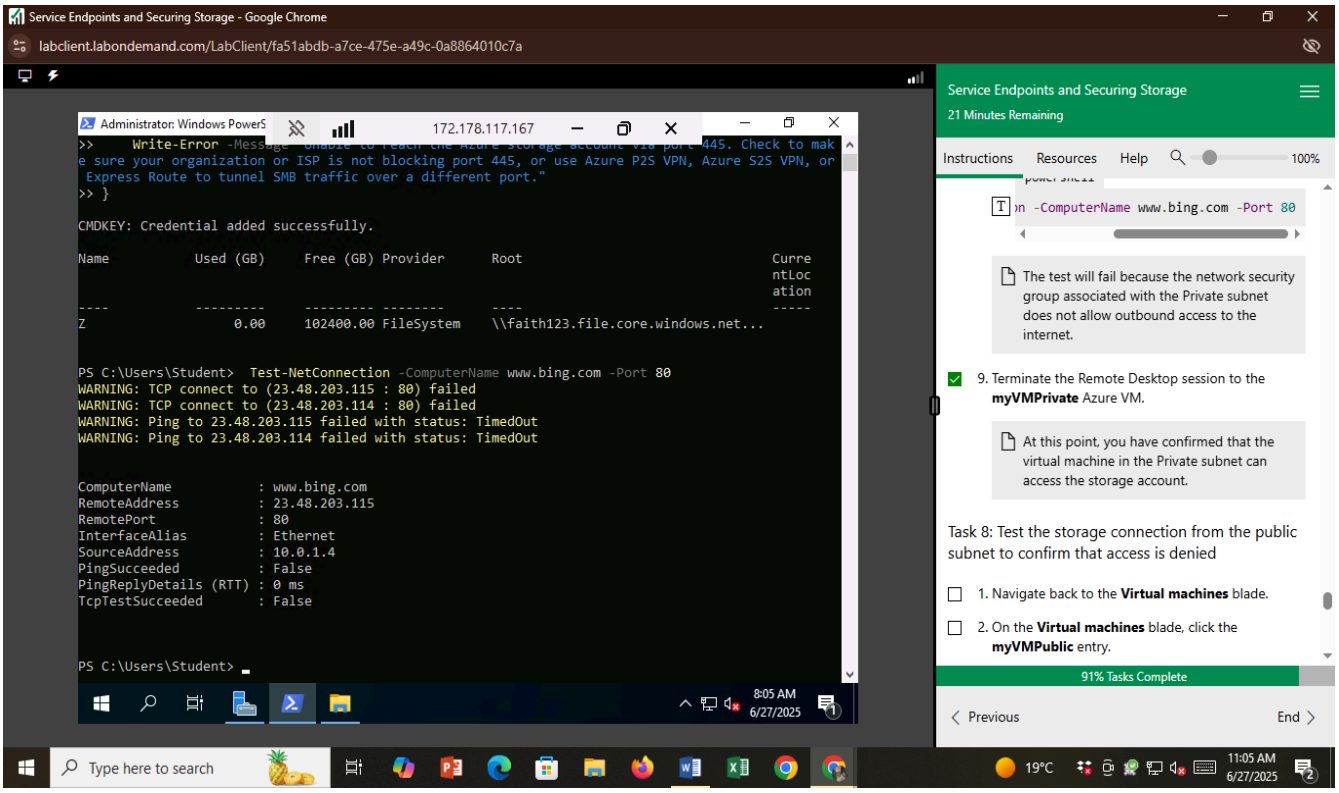
Step 7

I started File Explorer and verified that the Z: drive mapping has been successfully created.Here is my screenshot



Step 8

Next, from the console pane of the Windows PowerShell ISE console, I ran the command: Test-NetConnection -ComputerName www.bing.com -Port 80 to verify that the virtual machine has no outbound connectivity to the internet.The test failed because the network security group associated with the Private subnet does not allow outbound access to the internet.Here is my screenshot



Step 9

I terminated the Remote Desktop session to the myVMPrivate Azure VM.At this point,I have confirmed that the virtual machine in the Private subnet can access the storage account.

TASK 8:TEST THE STORAGE CONNECTION FROM THE PUBLIC SUBNET TO CONFIRM THAT ACCESS IS DENIED

Step 1

I navigate back to the Virtual machines blade.

Step 2

On the Virtual machines blade, I clicked the myVMPublic entry.

Step 3

On the myVMPublic blade, I clicked Connect and, in the drop down menu, clicked Connect.

Step 4

I clicked Connect via RDP and used it to connect to the myVMPublic Azure VM via Remote Desktop.

Step 5

Within the Remote Desktop session to myVMPublic, I clicked Start and then Windows PowerShell ISE.

Step 6

Within the Windows PowerShell ISE window, I opened the Script pane, then pasted and ran the same PowerShell script that I ran within the Remote Desktop session to the myVMPrivate Azure VM.

Step 7

Next, from the console pane of the Windows PowerShell ISE console, I ran the command:Test-NetConnection -ComputerName www.bing.com -Port 80 to verify that the virtual machine has outbound connectivity to the internet.The test succeeded because there is no outbound security rule to deny internet on the Public subnet.

Step 8

I terminated the Remote Desktop session to the myVMPublic Azure VM.At this point, I have confirmed that the virtual machine in the Public subnet cannot access the storage account, but has access to the internet.

I cleaned up my resources and completed the lab.